

Julin Shaji | Curriculum Vitae

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Summary

- Hardware Verification Engineer with a solid background in interactive theorem proving and model checking.
- Experienced in applying formal methods to verify both software and hardware systems.
- Seeking to leverage expertise in formal verification to build provably correct, reliable hardware designs.

Education

Master of Science (by research), Computer Science & Engineering

Indian Institute of Technology, Palakkad

India

2021–2026

- Produced provably correct hardware designs using type theory
- Generated hardware implementations of pattern matchers
- Intended use: Monitors for runtime verification
- Designs formally verified using Rocq theorem prover

CGPA: 8.2/10

Electives taken: Synthesis of digital systems, Programs-proofs-types, Logic, Functional programming

Publication: Certified hardware for regex matching (LOPSTR+PPDP @ ICFP 2026)

Bachelor of Technology, Computer Science & Engineering

Amal Jyothi College of Engineering, Kanjirappally

India

2014–2018

- Developed text summarisation tool using NLTK, NetworkX and PyQt5
- Created online auctioning platform using PHP as part of a team

CGPA: 7.73/10

Electives taken: Switching theory and logic design, Artificial Neural Networks, Artificial Intelligence

Projects

courant: MS research project

Verilog, Rocq, Clash

- Generation of provably correct monitors for use in runtime verification
- Hardware designs mathematically proven correct using Rocq theorem prover
- Tools involved: Vivado or Yosys-nextpnr, Rocq, Clash, tcl-bash scripts, Verilator
- Target FPGA: GW1N-9C (Gowin LittleBee) and xc7k70tfbv676 (Xilinx Kintex 7)
- Also worked with Verilator, yosys, Bluespec SystemVerilog, tcl

uvm-fifo: Synchronous FIFO tested with UVM v1.2

SystemVerilog

- Verified with SVA assertions
- Coverage testing performed
- Simulation with Questa 2025.2

clock: TUI clock in the style Qlocktwo clocks using minttea framework

OCaml

colorviews: Library to manipulate colour values

Python

- Supports viewing colours in RGB, HSL, HSV formats
- Also supports values with alpha for setting transparency level
- Values can be made from their numeric values, hexcode or CSS3 color keywords

calsh: Script to make the cal command of Linux interactive

bash

Work experience

Indian Institute of Technology, Palakkad

Teaching assistant

Palakkad

2021–2023

- Organized practical programming sessions
- Concepts related to functional programming and compilers
- Involved use of sml, Haskell and C

Talent Development, Tata Consultancy Services

Chennai
2020–2021

System Engineer

- Developed chatbots using Rasa framework
- Worked on NLP using spaCy
- Curated material on advanced Python and asyncio for internal MOOC platform
- Helped organize hackathons for other employees

Digital Talent Development, Tata Consultancy Services

Chennai
2018–2020

Assistant System Engineer

- Worked with time series data using pandas and R (dplyr, lubridate, rmarkdown)
- Scraped web data for analysis. Used BeautifulSoup for data cleaning.
- Curated material on Lua, Microsoft bot framework for internal MOOC platform
- Made tiny web apps using Python Flask

Tata Consultancy Services

Kochi (remote)
2018

Intern

- Created a dynamic web application using React.js
- Involved use of TypeScript, express.js and Google firebase
- Used MongoDB and mongoose for local tests

Computer skills

ITPs: Rocq, Agda, Lean, Twelf, Isabelle/HOL

OS: Linux, BSD

Others: UVM, PSL

Languages: OCaml, Haskell, sml, Python

Editors: emacs, vim, vscode

HDLs: Clash, Bluespec, Verilog/SystemVerilog

Scripting: Bash, Perl, Tcl

Events attended

Quantum circuits and quantum algorithms

Solan, India

Summer school, Organized by ACM India and National Quantum Mission

2025

Oregon Programming Languages Summer School 2024

Boston, USA

Received grant

2024

12th Summer School on Formal Techniques

Menlo Park, USA

Online participant

2023

Languages known

English, Malayalam: Fluent

Hindi, Tamil: Basic

French: Basic

Hobbies

Open source hardware design, Type theory

Graphic design, reading, travel

Exploring different cultures and languages

Achievements and activities

5000+ points on StackOverflow

Open source contributions: qutebrowser, vulture, typeshed

Speaker at several hobby conferences. Including PyCons, FOSDEM, LibrePlanet, MiniDebconf

Active in open source software communities, volunteer at several events

Attended programming language research events like POPL, PLDI, SPLASH and FSTTCS